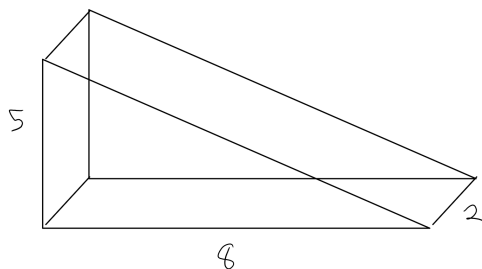
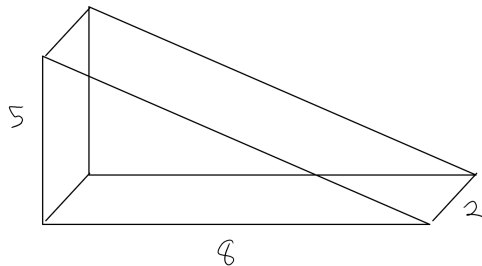
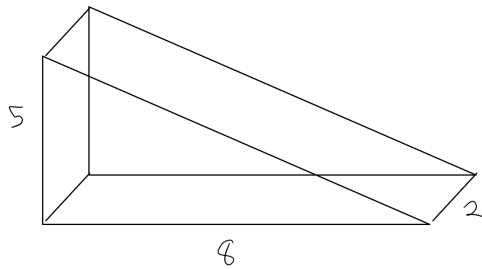


Volume via cross section

Goal. Use integration to find the volume of various solids.

1. Set up three different integrals to find the volume of concrete in the ramp drawn below (where the dimensions are given in meters). Draw a picture to correspond to each integral. Set up your axes and be sure to draw an arbitrary cross-section (at some fixed x , y , or z). In one of your pictures, the cross-section should be a triangle, and in the others, a rectangle. Evaluate the integrals to see that you obtain the same answer. Hint: When the cross-section is a rectangle in this example, its area will depend on the slope of the hypotenuse of the triangle.



2. Draw the solid whose base is the triangle enclosed by the line $x + y = 3$, the y -axis, and the x -axis, and whose cross-sections perpendicular to the y -axis are squares. Then use an integral to find its volume.

- 3.
- Object A has a base bounded by the circle $x^2 + y^2 = 4$, and its cross-sections perpendicular to the y -axis are squares.
 - Object B has a base bounded by the function $y = 4 - x^2$ and the x -axis, and its cross-sections perpendicular to the x -axis are rectangles with height 4.

Explain how we know the volumes of objects A and B are equal without computing the volumes.